$k = 4$ in characteristic 2 past one word: IC falls from 3.6× to 0.45× of rho's theoretical cost over $m = 15$–18 (clean sessions, single targets); a crossover is indicated, not established

2026-10-15 to 2026-11-10. `research/gaudry_k4_char2_crossover_20261015/`. Protocol registered (commit `e403626cb`) before any claim run. Stacked on 1883 **Class: engineering** (Parts D and F) and a **comparison against rho on non-standard curves** (Part E).

What finished

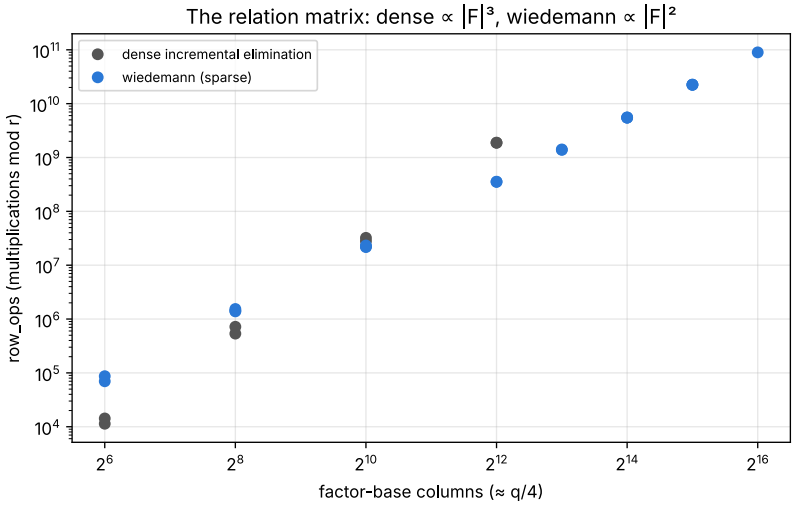
The container restarted many times during the claim runs and each restart kills every run in flight (`ecbench` cannot resume a session). The first attempts, on the registered binary (`c143cf62…`), were killed; they are kept as `*.attemptN-killed` and **no number below uses them**. The remaining runs completed on a second binary (`ad1dd91a…`) that adds checkpoint/resume (a run logs its oracle calls and resumes by replaying the walk against the log; unit tests compare every counter with an uninterrupted run). **Clean, complete sessions:** Part D at $m = 8, 10, 12, 14$ (21/21), Part E at $m = 15$ (6/6), $m = 16$ (4/4), $m = 17$ (IC 2/2 in two sessions, rho 2/2 and 1/1), $m = 18$ IC (1/1), Part F (15/15). All pass the integrity audit (`verify --replay 0`). **Not done:** the $m = 18$ rho run was still running when this was written; Part G (isolated hardware: no isolab credentials in this container); **no session of this study has been replayed from scratch** (hours to days each, against a restart every few hours). The sessions on the second binary rest on the tested equivalence, not on a replay. The $m = 18$ IC session resumed after a filesystem fault (see EXPERIMENTS W33). The two replayed sessions are 1883's, re-executed with the first binary (27/27 reproduced).

1. The matrix was the obstacle

In 1883 the dense matrix's `row_ops` grew as $|F|^3$ and would have reached rho's cost near $m = 18$. The `wiedemann` matrix (singleton-filtered core, Wiedemann on $R^T D R$, Berlekamp–Massey, exact check) grows as $|F|^2$:

m	columns	dense `row_ops`	sparse `row_ops`	dense ÷ sparse	trials, sparse ÷ dense
8	62	11,370; 14,252	69,871; 86,869	0.16; 0.16	1.0000
10	262	718,813; 536,268	1,536,771; 1,423,931	0.47; 0.38	1.0000
12	1,045	32.2 M; 28.0 M	22.2 M; 21.7 M	**1.45; 1.29**	1.0008; 1.0040
14	4,123	1.88 B (Part D), 1.91 B (1883)	0.356 B (Part D), 0.353 B (Part F)	**5.29** (paired); 5.4	1.0006; 1.0003

The sparse solver loses where the matrices are tiny and wins from $m = 12$. Its share of the IC cost is 0.36 %, 0.68 %, 1.36 % and 2.57 % at $m = 15, 16, 17, 18$, doubling per m (5 % at $m = 19$). The $m = 14$ row is now paired (one workload).



`row_ops` against factor-base columns. `raw/summary.json`.

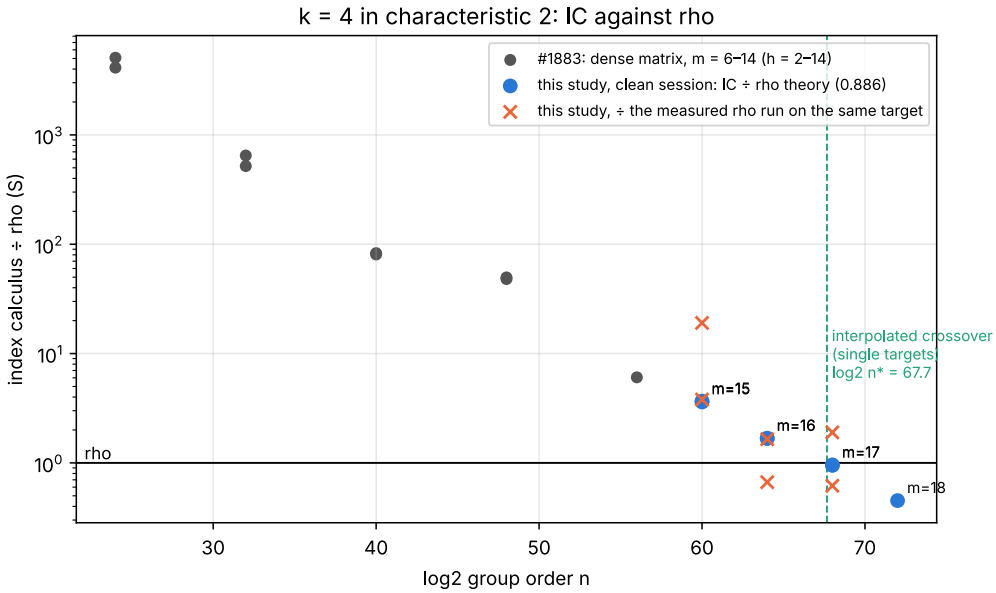
2. IC against rho, cofactor $h = 2$, $m = 15$–18

m	$\log_2 n$	target	IC S	rho S	IC ÷ rho theory (0.886)	IC ÷ rho run	matrix share
15	60	`W588ac37c7439`	3.198	0.168	3.61	19.1	0.36 %
15	60	`W761fd6d9429f`	3.253	0.856	3.67	3.8	0.36 %
16	64	`Wa1b3b10e6968`	1.482	0.899	1.67	1.65	0.68 %
16	64	`W37a8388076b0`	1.490	2.235	1.68	0.67	0.68 %
17	68	`Wd5f7dba84d0e`	0.842	1.361	0.95	0.62	1.36 %

17	68	`W591d11cef5d4`	0.849	0.447	0.96	1.90	1.36 %
18	72	`We3b266046ebb`	0.400	*running*	**0.45**	–	2.57 %

IC used 190–196 trials per relation at every size. The two targets of each size agree within 1 % in IC's cost. The fit of $\log_2$(IC ÷ rho theory) against $\log_2 n$ has slope -0.246 ($-\frac{1}{4}$ expected) and the interpolation crosses 1 between $m = 16$ and 17, at $\log_2 n^* \approx 67.7$ ($m^* \approx 16.9$), against 1883's extrapolated $2^{68.1}$. **Measured:** IC's cost relative to rho's *theoretical* cost falls steadily from 3.6× to 0.45× over $m = 15$–18. **Not shown:** that IC beats rho in practice. The $m = 18$ point is one target, individual rho runs scatter about 5× around their expectation (IC ÷ rho run spans 0.62–19 at $m = 15$–17), and the rho half of $m = 18$ is unfinished.

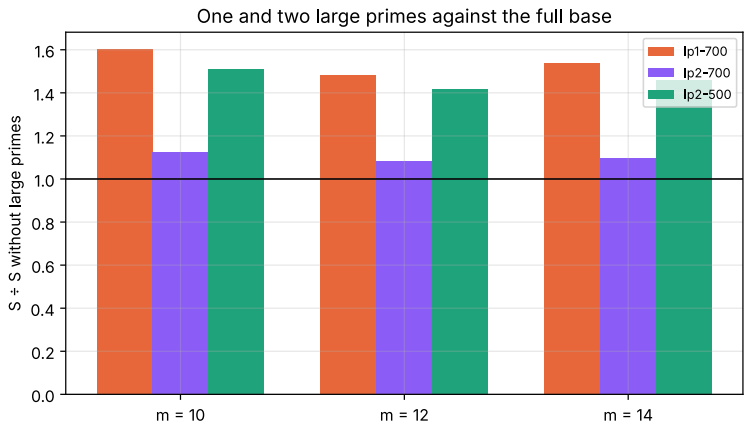
Open (relayed from an independent red-team review of this study, not re-verified here: its report was said to be on branch `claude/k4-char2-redteam-20261026`, which I could not fetch): the unit price (corrections of 0.3×–3.1× are reported and unresolved), and every `rho.negation` row is a lower bound (canonicalisations are uncharged, so rho's true cost is higher, which favours IC). The review projects $m^* \approx 17.6$ (16.3–19.2) and judges the crossover not established; this study agrees that two targets per size, one at $m = 18$, at an unresolved unit price, do not establish it.



IC ÷ rho against $\log_2 n$: 1883 (grey, dense matrix, $h = 2$–14), this study's clean-session points (filled), IC ÷ the measured rho run (crosses). `raw/summary.json`.

3. Large primes lose at these sizes (Part F, clean)

S ÷ the same run without large primes: one large prime over 70 % of the columns 1.60, 1.48, 1.54; two large primes over 70 %: 1.13, 1.08, 1.10; over 50 %: 1.51, 1.42, 1.46 ($m = 10, 12, 14$). The matrix they shrink is under 1 % of the cost (its cost falls to 0.52–0.78×); merging and the extra trials cost more. By the protocol's model the variant pays near $n \approx 2^{92}$. A recorded negative at every reachable size.



$S \div S$ without large primes. `raw/summary.json`.

Predictions and checks

D1 *not met as registered* (a spec error: the measurement seed differs from 1883′s, so the dense arms are fresh samples; the pilot with the seed reused matched exactly, and the first binary replays all 27 committed 1883 runs). D2 (5.29× at $m = 14$, paired), D3, E2 (all four sizes), E4 (2.57 %), F1, F2: held. E3: IC below rho's theory at $m = 18$ (0.45×); the measured-run half is pending. E5: the interpolation gives 16.9 (single targets, unresolved unit price). Frobenius parity: even on 375/375 curves ($m = 4$–18), measured, not proved.

Scope. Curves over $\mathbb{F}_{2^{4m}}$ are not standard curves. Next: the $m = 18$ rho run, a second $m = 18$ target, and from-scratch replays of the sessions on the second binary.

Environment: Claude Code cloud container, Xeon @ 2.10 GHz, 4 vCPU, four sessions at a time on separate cores (L1). Conductor task T-4.